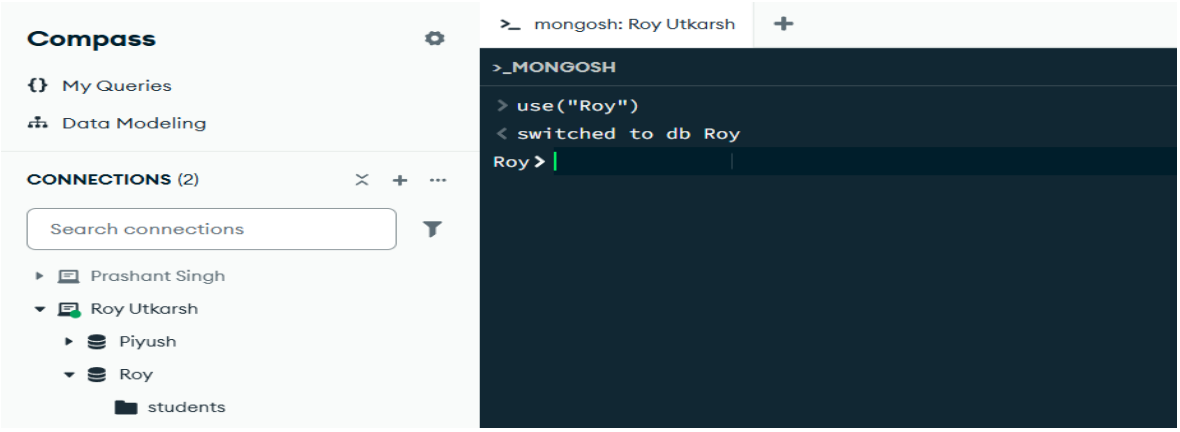
1. CREATE DATABASE

MongoDB Query:

use Students

Explanation: Creates or switches to the Student database.

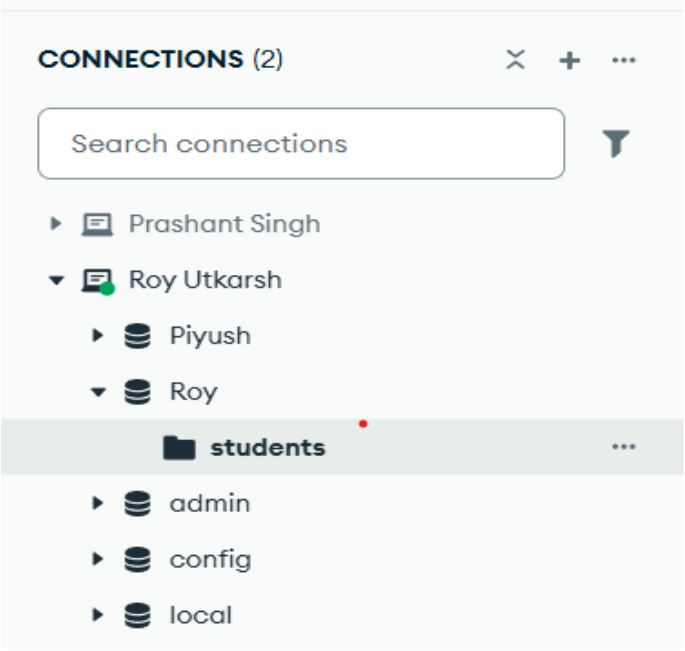


2. CREATE COLLECTION

MongoDB Query:

db. create Collection("students")

Explanation: Creates a collection named students.



3. INSERT 50 STUDENTS

MongoDB Query:

```
db. students. Insert Many([
  { rollNo: 1, name: "Aarav Patel", age: 19, marks: 78, city: "Ahmedabad" },
  { rollNo: 2, name: "Diya Shah", age: 20, marks: 85, city: "Surat" },
  { rollNo: 3, name: "Vivaan Mehta", age: 19, marks: 72, city: "Vadodara" },
  { rollNo: 4, name: "Anaya Desai", age: 18, marks: 91, city: "Rajkot" },
  { rollNo: 5, name: "Arjun Patel", age: 20, marks: 68, city: "Gandhinagar" },
```

```
  { rollNo: 6, name: "Kiara Shah", age: 19, marks: 88, city: "Bhavnagar" },
  { rollNo: 7, name: "Reyansh Joshi", age: 18, marks: 76, city: "Jamnagar" },
  { rollNo: 8, name: "Myra Mehta", age: 20, marks: 94, city: "Ahmedabad" },
  { rollNo: 9, name: "Aditya Desai", age: 19, marks: 81, city: "Surat" },
  { rollNo: 10, name: "Riya Patel", age: 18, marks: 73, city: "Vadodara" },

  { rollNo: 11, name: "Krish Shah", age: 20, marks: 89, city: "Rajkot" },
  { rollNo: 12, name: "Aadhya Joshi", age: 19, marks: 95, city: "Gandhinagar" },
  { rollNo: 13, name: "Dhruv Patel", age: 18, marks: 67, city: "Bhavnagar" },
  { rollNo: 14, name: "Ishita Desai", age: 20, marks: 82, city: "Jamnagar" },
  { rollNo: 15, name: "Yash Mehta", age: 19, marks: 74, city: "Ahmedabad" },
  { rollNo: 16, name: "Avni Shah", age: 18, marks: 90, city: "Surat" },
  { rollNo: 17, name: "Harsh Patel", age: 20, marks: 79, city: "Vadodara" },
  { rollNo: 18, name: "Navya Desai", age: 19, marks: 87, city: "Rajkot" },
  { rollNo: 19, name: "Rudra Joshi", age: 18, marks: 71, city: "Gandhinagar" },
  { rollNo: 20, name: "Meera Patel", age: 20, marks: 93, city: "Bhavnagar" },

  { rollNo: 21, name: "Kunal Shah", age: 19, marks: 65, city: "Jamnagar" },
  { rollNo: 22, name: "Isha Mehta", age: 18, marks: 84, city: "Ahmedabad" },
  { rollNo: 23, name: "Manav Patel", age: 20, marks: 77, city: "Surat" },
  { rollNo: 24, name: "Saanvi Desai", age: 19, marks: 92, city: "Vadodara" },
  { rollNo: 25, name: "Dev Joshi", age: 18, marks: 69, city: "Rajkot" },
  { rollNo: 26, name: "Kavya Shah", age: 20, marks: 86, city: "Gandhinagar" },
  { rollNo: 27, name: "Rohan Patel", age: 19, marks: 75, city: "Bhavnagar" },
  { rollNo: 28, name: "Tanya Mehta", age: 18, marks: 89, city: "Jamnagar" },
  { rollNo: 29, name: "Nirav Desai", age: 20, marks: 80, city: "Ahmedabad" },
  { rollNo: 30, name: "Pooja Patel", age: 19, marks: 96, city: "Surat" },
```

```
  { rollNo: 31, name: "Siddharth Shah", age: 18, marks: 70, city: "Vadodara" },
  { rollNo: 32, name: "Nisha Joshi", age: 20, marks: 83, city: "Rajkot" },
  { rollNo: 33, name: "Mihir Patel", age: 19, marks: 88, city: "Gandhinagar" },
  { rollNo: 34, name: "Jhanvi Desai", age: 18, marks: 78, city: "Bhavnagar" },
  { rollNo: 35, name: "Parth Mehta", age: 20, marks: 91, city: "Jamnagar" },
  { rollNo: 36, name: "Khushi Shah", age: 19, marks: 74, city: "Ahmedabad" },
  { rollNo: 37, name: "Darsh Patel", age: 18, marks: 85, city: "Surat" },
  { rollNo: 38, name: "Riddhi Joshi", age: 20, marks: 97, city: "Vadodara" },
  { rollNo: 39, name: "Meet Desai", age: 19, marks: 66, city: "Rajkot" },
  { rollNo: 40, name: "Mansi Patel", age: 18, marks: 81, city: "Gandhinagar" },

  { rollNo: 41, name: "Jay Shah", age: 20, marks: 79, city: "Bhavnagar" },
  { rollNo: 42, name: "Hetal Mehta", age: 19, marks: 90, city: "Jamnagar" },
  { rollNo: 43, name: "Akash Patel", age: 18, marks: 73, city: "Ahmedabad" },
  { rollNo: 44, name: "Pallavi Desai", age: 20, marks: 87, city: "Surat" },
  { rollNo: 45, name: "Raj Joshi", age: 19, marks: 82, city: "Vadodara" },
  { rollNo: 46, name: "Simran Shah", age: 18, marks: 94, city: "Rajkot" },
  { rollNo: 47, name: "Sahil Patel", age: 20, marks: 68, city: "Gandhinagar" },
  { rollNo: 48, name: "Neha Mehta", age: 19, marks: 76, city: "Bhavnagar" },
  { rollNo: 49, name: "Vatsal Desai", age: 18, marks: 89, city: "Jamnagar" },
  { rollNo: 50, name: "Palak Joshi", age: 20, marks: 93, city: "Ahmedabad" }
])
```

```
> db.students.insertMany([
    { rollNo: 1, name: "Aarav Patel", age: 19, marks: 78, city: "Ahmedabad" },
    { rollNo: 2, name: "Diya Shah", age: 20, marks: 85, city: "Surat" },
    { rollNo: 3, name: "Vivaan Mehta", age: 19, marks: 72, city: "Vadodara" },
    { rollNo: 4, name: "Anaya Desai", age: 18, marks: 91, city: "Rajkot" },
    { rollNo: 5, name: "Arjun Patel", age: 20, marks: 68, city: "Gandhinagar" },
    { rollNo: 6, name: "Kiara Shah", age: 19, marks: 88, city: "Bhavnagar" },
    { rollNo: 7, name: "Reyansh Joshi", age: 18, marks: 76, city: "Jamnagar" },
    { rollNo: 8, name: "Myra Mehta", age: 20, marks: 94, city: "Ahmedabad" },
    { rollNo: 9, name: "Aditya Desai", age: 19, marks: 81, city: "Surat" },
    { rollNo: 10, name: "Riya Patel", age: 18, marks: 73, city: "Vadodara" },

    { rollNo: 11, name: "Krish Shah", age: 20, marks: 89, city: "Rajkot" },
    { rollNo: 12, name: "Aadhya Joshi", age: 19, marks: 95, city: "Gandhinagar" },
    { rollNo: 13, name: "Dhruv Patel", age: 18, marks: 67, city: "Bhavnagar" },
    { rollNo: 14, name: "Ishita Desai", age: 20, marks: 82, city: "Jamnagar" },
    { rollNo: 15, name: "Yash Mehta", age: 19, marks: 74, city: "Ahmedabad" },
    { rollNo: 16, name: "Avni Shah", age: 18, marks: 90, city: "Surat" },
    { rollNo: 17, name: "Harsh Patel", age: 20, marks: 79, city: "Vadodara" },
    { rollNo: 18, name: "Navya Desai", age: 19, marks: 87, city: "Rajkot" },
    { rollNo: 19, name: "Rudra Joshi", age: 18, marks: 71, city: "Gandhinagar" },
    { rollNo: 20, name: "Meera Patel", age: 20, marks: 93, city: "Bhavnagar" },

    { rollNo: 21, name: "Kunal Shah", age: 19, marks: 65, city: "Jamnagar" },
    { rollNo: 22, name: "Isha Mehta", age: 18, marks: 84, city: "Ahmedabad" },
    { rollNo: 23, name: "Manav Patel", age: 20, marks: 77, city: "Surat" },
    { rollNo: 24, name: "Saanvi Desai", age: 19, marks: 92, city: "Vadodara" },
```

Explanation: Inserts multiple student documents into the collection

4. DISPLAY ALL STUDENTS

MongoDB Query:

db. students. Find ()

Explanation: Retrieves documents from the students collection.

```
> db.students.find()
< {
    _id: ObjectId('6ab7c742350d31497edea7c8'),
    rollNo: 1,
    name: 'Aarav Patel',
    age: 19,
    marks: 78,
    city: 'Ahmedabad'
  }
  {
    _id: ObjectId('6ab7c742350d31497edea7c9'),
    rollNo: 2,
    name: 'Diya Shah',
    age: 20,
    marks: 85,
    city: 'Surat'
  }
  {
    _id: ObjectId('6ab7c742350d31497edea7ca'),
    rollNo: 3,
    name: 'Vivaan Mehta',
    age: 19,
    marks: 72,
    city: 'Vadodara'
  }
  {
```

5. EQUAL TO — $eq

MongoDB Query:

db. students. Find ({marks: {$eq: 88}})

Explanation: Finds students whose marks are exactly 88

```
> db.students.find({marks: {$eq: 88}})
< {
    _id: ObjectId('6ab7c742350d31497edea7cd'),
    rollNo: 6,
    name: 'Kiara Shah',
    age: 19,
    marks: 88,
    city: 'Bhavnagar'
  }
  {
    _id: ObjectId('6ab7c742350d31497edea7e8'),
    rollNo: 33,
    name: 'Mihir Patel',
    age: 19,
    marks: 88,
    city: 'Gandhinagar'
  }
```

6. GREATER THAN – $gt

MongoDB Query:

db.students.find({marks: {$gt: 85}})

Explanation: Finds students whose marks are greater than 85



7. LESS THAN – $lt

MongoDB Query:

db. students. find ({marks: {$lt: 70}})

Explanation: Finds students whose marks are less than 70.

```
> db.students.find({marks: {$lt: 70}})
< {
    _id: ObjectId('6ab7c742350d31497edea7cc'),
    rollNo: 5,
    name: 'Arjun Patel',
    age: 20,
    marks: 68,
    city: 'Gandhinagar'
  }
  {
    _id: ObjectId('6ab7c742350d31497edea7d4'),
    rollNo: 13,
    name: 'Dhruv Patel',
    age: 18,
    marks: 67,
    city: 'Bhavnagar'
  }
  {
    _id: ObjectId('6ab7c742350d31497edea7dc'),
    rollNo: 21,
    name: 'Kunal Shah',
    age: 19,
    marks: 65,
    city: 'Jamnagar'
  }
  {
```

8. GREATER THAN OR EQUAL TO — $gte

MongoDB Query:

db. students. find({marks: {$gte: 90}})

Explanation: Finds students whose marks are 90 or above.

```
> db.students.find({marks: {$gte: 90}})
< {
    _id: ObjectId('6ab7c742350d31497edea7cb'),
    rollNo: 4,
    name: 'Anaya Desai',
    age: 18,
    marks: 91,
    city: 'Rajkot'
  }
  {
    _id: ObjectId('6ab7c742350d31497edea7cf'),
    rollNo: 8,
    name: 'Myra Mehta',
    age: 20,
    marks: 94,
    city: 'Ahmedabad'
  }
  {
    _id: ObjectId('6ab7c742350d31497edea7d3'),
    rollNo: 12,
    name: 'Aadhya Joshi',
    age: 19,
    marks: 95,
    city: 'Gandhinagar'
  }
  {
```

9. LESS THAN OR EQUAL TO — $lte

MongoDB Query:

db. students. find({marks: {$lte: 70}})

Explanation: Finds students whose marks are 70 or below.

```
> db.students.find({marks: {$lte: 70}})
< {
    _id: ObjectId('6ab7c742350d31497edea7cc'),
    rollNo: 5,
    name: 'Arjun Patel',
    age: 20,
    marks: 68,
    city: 'Gandhinagar'
  }
  {
    _id: ObjectId('6ab7c742350d31497edea7d4'),
    rollNo: 13,
    name: 'Dhruv Patel',
    age: 18,
    marks: 67,
    city: 'Bhavnagar'
  }
  {
    _id: ObjectId('6ab7c742350d31497edea7dc'),
    rollNo: 21,
    name: 'Kunal Shah',
    age: 19,
    marks: 65,
    city: 'Jamnagar'
  }
  {
```

10. UPDATE OPERATION

MongoDB Query:

```
db.students.updateOne(
  {roll:5},
  {$set:{marks:95}}
)
```

Explanation: Updates the marks to 95 of the student with roll number 5.

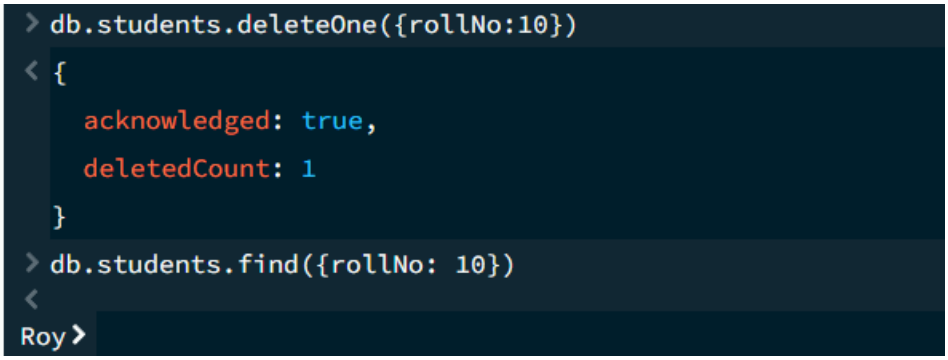
```
> db.students.updateOne(
    {rollNo: 5},
    {$set:{marks:95}}
  )
< {
    acknowledged: true,
    insertedId: null,
    matchedCount: 1,
    modifiedCount: 1,
    upsertedCount: 0
  }
> db.students.find({rollNo: 5})
< {
    _id: ObjectId('6ab7c742350d31497edea7cc'),
    rollNo: 5,
    name: 'Arjun Patel',
    age: 20,
    marks: 95,
    city: 'Gandhinagar'
  }
```

11. DELETE ONE OPERATION

MongoDB Query:

```
db.students.deleteOne({roll:10})
```

Explanation: Deletes one document matching roll number 10



12. DELETE MANY OPERATION

MongoDB Query:

```
db.students.deleteMany({
  marks: {$lt:60}
})
```

Explanation: Deletes all students having marks below 60.

```
> db.students.deleteMany({
    marks: {$lt:60}
  })
< {
    acknowledged: true,
    deletedCount: 0
  }
```

13. AND OPERATION – $and

MongoDB Query:

```
db. students. find({
  $and:[
    {marks:{$gt:80}},
    {age:19}
  ]
})
```

Explanation: Finds students satisfying both conditions.

```
> db.students.find({
    $and:[
      {marks:{$gt:80}},
      {age:19}
    ]
  })
< {
    _id: ObjectId('6ab7c742350d31497edea7cd'),
    rollNo: 6,
    name: 'Kiara Shah',
    age: 19,
    marks: 88,
    city: 'Bhavnagar'
  }
  {
    _id: ObjectId('6ab7c742350d31497edea7d0'),
    rollNo: 9,
    name: 'Aditya Desai',
    age: 19,
    marks: 81,
    city: 'Surat'
  }
  {
    _id: ObjectId('6ab7c742350d31497edea7d3'),
    rollNo: 12,
    name: 'Aadhya Joshi',
```

14. OR OPERATION — $or

MongoDB Query:

```
db.students.find({
  $or:[
    {city:"Lucknow"},
    {city:"Kanpur"}
  ]
})
```

Explanation: Finds students from Lucknow or Kanpur.

```
>_MONGOSH
> db.students.find({
    $or:[
      {city:"Surat"},
      {city:"Gandhinagar"}
    ]
  })
< {
    _id: ObjectId('6ab7c742350d31497edea7c9'),
    rollNo: 2,
    name: 'Diya Shah',
    age: 20,
    marks: 85,
    city: 'Surat'
  }
  {
    _id: ObjectId('6ab7c742350d31497edea7cc'),
    rollNo: 5,
    name: 'Arjun Patel',
    age: 20,
    marks: 95,
    city: 'Gandhinagar'
  }
  {
    _id: ObjectId('6ab7c742350d31497edea7d0'),
    rollNo: 9,
    name: 'Aditya Desai',
    age: 19,
    marks: 81,
```

15. SORT OPERATION

A.MongoDB Query:

db. students. find (). sort({marks : 1})

Explanation: Sorts students by marks from lowest to highest. Use 1 for ascending order.

```
> db.students.find().sort({marks : 1})
< {
    _id: ObjectId('6ab7c742350d31497edea7dc'),
    rollNo: 21,
    name: 'Kunal Shah',
    age: 19,
    marks: 65,
    city: 'Jamnagar'
  }
  {
    _id: ObjectId('6ab7c742350d31497edea7ee'),
    rollNo: 39,
    name: 'Meet Desai',
    age: 19,
    marks: 66,
    city: 'Rajkot'
  }
  {
    _id: ObjectId('6ab7c742350d31497edea7d4'),
    rollNo: 13,
    name: 'Dhruv Patel',
    age: 18,
    marks: 67,
    city: 'Bhavnagar'
  }
  {
```

B. MongoDB Query:

db. students. find (). Sort ({marks: -1})

Explanation: Sorts students by marks from highest to lowest. Use -1 for descending order.

```
> db.students.find().sort({marks : -1})
< {
    _id: ObjectId('6ab7c742350d31497edea7ed'),
    rollNo: 38,
    name: 'Riddhi Joshi',
    age: 20,
    marks: 97,
    city: 'Vadodara'
  }
  {
    _id: ObjectId('6ab7c742350d31497edea7e5'),
    rollNo: 30,
    name: 'Pooja Patel',
    age: 19,
    marks: 96,
    city: 'Surat'
  }
  {
    _id: ObjectId('6ab7c742350d31497edea7cc'),
    rollNo: 5,
    name: 'Arjun Patel',
    age: 20,
    marks: 95,
    city: 'Gandhinagar'
  }
  {
```

16. LIMT OPERATION

MongoDB Query:

db. students. find (). limit (5)

Explanation: Displays only the first five documents.

```
> db.students.find().limit(5)
< {
    _id: ObjectId('6ab7c742350d31497edea7c8'),
    rollNo: 1,
    name: 'Aarav Patel',
    age: 19,
    marks: 78,
    city: 'Ahmedabad'
  }
  {
    _id: ObjectId('6ab7c742350d31497edea7c9'),
    rollNo: 2,
    name: 'Diya Shah',
    age: 20,
    marks: 85,
    city: 'Surat'
  }
  {
    _id: ObjectId('6ab7c742350d31497edea7ca'),
    rollNo: 3,
    name: 'Vivaan Mehta',
    age: 19,
    marks: 72,
    city: 'Vadodara'
  }
  {
    _id: ObjectId('6ab7c742350d31497edea7cb'),
```

17. SEARCH BY CITY

MongoDB Query:

db.students.find({city:"Lucknow"})

Explanation: Finds all students whose city is Lucknow.

```
> db.students.find({city:"Surat"})
< {
    _id: ObjectId('6ab7c742350d31497edea7c9'),
    rollNo: 2,
    name: 'Diya Shah',
    age: 20,
    marks: 85,
    city: 'Surat'
  }
  {
    _id: ObjectId('6ab7c742350d31497edea7d0'),
    rollNo: 9,
    name: 'Aditya Desai',
    age: 19,
    marks: 81,
    city: 'Surat'
  }
  {
    _id: ObjectId('6ab7c742350d31497edea7d7'),
    rollNo: 16,
    name: 'Avni Shah',
    age: 18,
    marks: 90,
    city: 'Surat'
  }
  {
    _id: ObjectId('6ab7c742350d31497edea7de'),
    rollNo: 23,
```

18. COUNT DOCUMENTS

MongoDB Query:

db. students. count Documents ()

Explanation: Counts documents currently present in the collection (performing after the delete function).

